

10 2025-02-04 | Week 04 | Lecture 10

This section is based on sections 2.4 in the textbook.

10.1 Conditional Probability: Examples

For any two events A and B , conditional probability describes the probability that A happens given that we know B happened.

Definition 33 (Conditional Probability). Let A, B be events, and assume that $\mathbb{P}[A] > 0$. Then the **conditional probability of B , given A** , denoted $\mathbb{P}[B | A]$, is given by the formula

$$\mathbb{P}[B | A] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[A]}.$$

Example 34 (Conditional Probability). I roll two dice and add them up. Call this number X . Then $\mathbb{P}[X = 7] = \frac{1}{6}$, which we can see by counting up the possibilities. Now, what about if I tell you that I rolled the dice and got a sum greater than 4? Now that you have a little additional information, you can exclude the possibilities that $X = 1, 2, 3$ or $X = 4$. So with this new information, you should evaluate the probability that $X = 7$ as greater than before. That is, now we want

$$\begin{aligned}\mathbb{P}[X = 7 | X > 4] &= \frac{\mathbb{P}[X = 7 \text{ and } X > 4]}{\mathbb{P}[X > 4]} \\ &= \frac{\mathbb{P}[X = 7]}{\mathbb{P}[X > 4]} \\ &= \frac{1/6}{30/36} \\ &= \frac{1}{5}.\end{aligned}$$

This answer makes sense because $\frac{1}{5} > \frac{1}{6}$, consistent with our intuition.

End of Example 34. $\square$

Example 35 (Urn - example of multiplication rule). An urn contains 6 white balls and 9 black balls. If 4 balls are drawn at random, what is the probability that the first 2 are white and the last 2 are black?

Solution: Let A be the event that the first two balls are white, and B the event that the last two balls are black. Then the desired probability is

$$\mathbb{P}[A \cap B] = \mathbb{P}[B | A] \mathbb{P}[A] \tag{2}$$

Now,

$$\mathbb{P}[A] = \frac{\binom{6}{2}}{\binom{15}{2}} = \frac{1}{7}.$$

Moreover, given that A occurs, then 4 white balls and 9 black balls remain. So

$$\mathbb{P}[B | A] = \frac{\binom{9}{2}}{\binom{13}{2}} = \frac{6}{13}.$$

Therefore by Eq. (2),

$$\mathbb{P}[A \cap B] = \frac{1}{7} \cdot \frac{6}{13} = \frac{6}{91} \approx 0.07$$

End of Example 35. $\square$

10.2 The law of total probability

Definition 36. We say that a collection of events $A_1, \dots, A_n$ is **exhaustive** if at least one must occur; that is,

$$A_1 \cup A_2 \cup \dots \cup A_n = S$$

where S is the sample space.

Theorem 37 (Law of Total Probability). *Let $A_1, \dots, A_n$ be events which are (1) mutually exclusive and (2) exhaustive. Then for any event B ,*

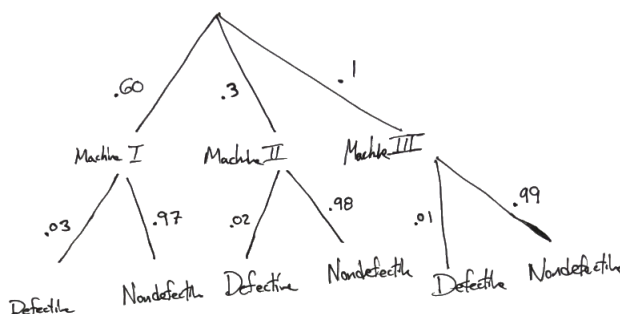
$$\mathbb{P}[B] = \sum_{k=1}^n \mathbb{P}[B | A_k] \mathbb{P}[A_k]$$

Example 38. A munitions factory produces 155mm artillery rounds using three machines: 60% of rounds are produced by machine I, 30% by machine II, and 10% by machine III. Engineers discover that

- 3% of rounds produced by machine I are defective
- 2% of rounds produced by machine II are defective
- 1% of rounds produced by machine III are defective.

Given a defective round, what is the probability that it was produced by machine III?

Solution: Let's make a diagram to illustrate this:



Let

D = [shell is defective]

M_1 = [shell was made by machine I]

M_2 = [shell was made by machine II]

M_3 = [shell was made by machine III].

We want to find $\mathbb{P}[M_3 | D] = \mathbb{P}[\text{machine III} | \text{defective}]$.

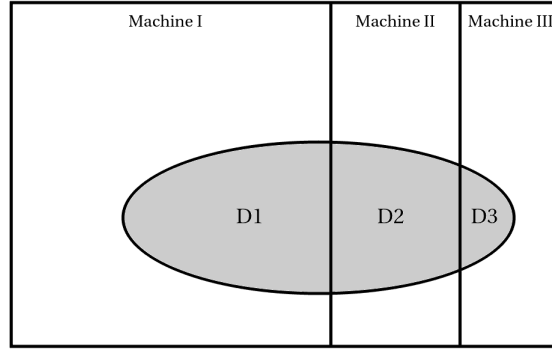
$$\mathbb{P}[M_3 | D] = \frac{\mathbb{P}[M_3 \cap D]}{\mathbb{P}[D]} \quad (3)$$

Now we compute the numerator and denominator separately:

By the multiplication rule,

$$\mathbb{P}[M_3 \cap D] = \mathbb{P}[D | M_3] \mathbb{P}[M_3] = (.01)(.1) = .001.$$

Next, we compute the denominator. The following figure illustrates the sample space, which consists of shells produced by machines I, II and III. The grey region represents the event that the randomly selected shell is defective.



$$\begin{aligned}
 \mathbb{P}[D] &= \mathbb{P}[D1] + \mathbb{P}[D2] + \mathbb{P}[D3] \\
 &= \mathbb{P}[D \cap M_1] + \mathbb{P}[D \cap M_2] + \mathbb{P}[D \cap M_3] \\
 &= \mathbb{P}[D | M_1] \mathbb{P}[M_1] + \mathbb{P}[D | M_2] \mathbb{P}[M_2] + \mathbb{P}[D | M_3] \mathbb{P}[M_3] \\
 &= (.03)(.6) + (.02)(.3) + (.01)(.1) \\
 &= .025
 \end{aligned}$$

Hence by Eq. (3),

$$\mathbb{P}[M_3 | D] = \frac{.001}{.025} = .04.$$

In other words, 4% of defective rounds are produce by machine III.

End of Example 38. $\square$

Example 39 (Conditional probability). Suppose you flip a coin 10 times and get more than 6 heads. What's the probability that you get less than 9 heads?

Solution: Let X be the number of heads. Let $A = [X < 9]$, and let $B = [X > 6]$. Then

$$\mathbb{P}[B] = \frac{\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{2^{10}}$$

Observe that $A \cap B = [X \in \{7, 8\}]$. Therefore

$$\mathbb{P}[A \cap B] = \frac{\binom{10}{7} + \binom{10}{8}}{2^{10}}$$

Therefore

$$\mathbb{P}[A | B] = \frac{\mathbb{P}[AB]}{\mathbb{P}[B]} = \frac{\binom{10}{7} + \binom{10}{8}}{\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}} = \frac{15}{16}.$$

End of Example 39. $\square$